Software Design Document (SDD)

EduTrack – Student Performance and Attendance Management System

---

1. Introduction

1.1 Purpose

The EduTrack system is designed to provide a digital platform for managing student academic records, attendance tracking, and performance monitoring.

The purpose of this system is to help educational institutions maintain organized records of student data while reducing manual work. EduTrack enables teachers and administrators to efficiently manage attendance, monitor student performance, and analyze academic progress through a centralized platform.

1.2 Scope

The EduTrack system provides the following features:

• User registration and authentication
• Student information management
• Attendance tracking system
• Student performance management
• Dashboard and analytics features

The system replaces traditional manual methods with a digital solution that improves efficiency, accuracy, and accessibility of academic records.

1.3 Problem Statement

Many educational institutions still rely on manual attendance registers and spreadsheets to track student records. This often results in errors, data inconsistency, and difficulty in generating reports.

The EduTrack system addresses these issues by providing an automated platform where student data, attendance, and performance records are securely stored and easily accessible.

---

2. System Requirements

2.1 Functional Requirements

The system shall support the following functionalities:

• Users shall be able to register and log in to the system.
• Administrators shall be able to manage student records.
• Teachers shall be able to record and update attendance.
• Teachers shall be able to record student performance and marks.
• Users shall be able to view attendance records.
• Users shall be able to view performance reports and statistics.

2.2 Non-Functional Requirements

The system must satisfy the following quality attributes:

• **Security:** The system must ensure secure authentication and authorization.
• **Performance:** The system should provide fast response times for user requests.
• **Scalability:** The system should support increasing numbers of students and records.
• **Reliability:** The system must ensure accurate and consistent data storage.
• **Maintainability:** The system architecture should allow easy updates and maintenance.

---

3. System Architecture

The EduTrack system follows a **three-tier architecture** consisting of presentation, application, and data layers.

Architecture Structure

Users (Admin / Teacher / Student)
↓
React Frontend
↓
Spring Boot Backend
↓
MySQL Database

Architecture Description

Frontend Layer

• The frontend is developed using React.js.
• It provides the user interface for interacting with the system.
• Users can view dashboards, manage students, record attendance, and monitor performance.

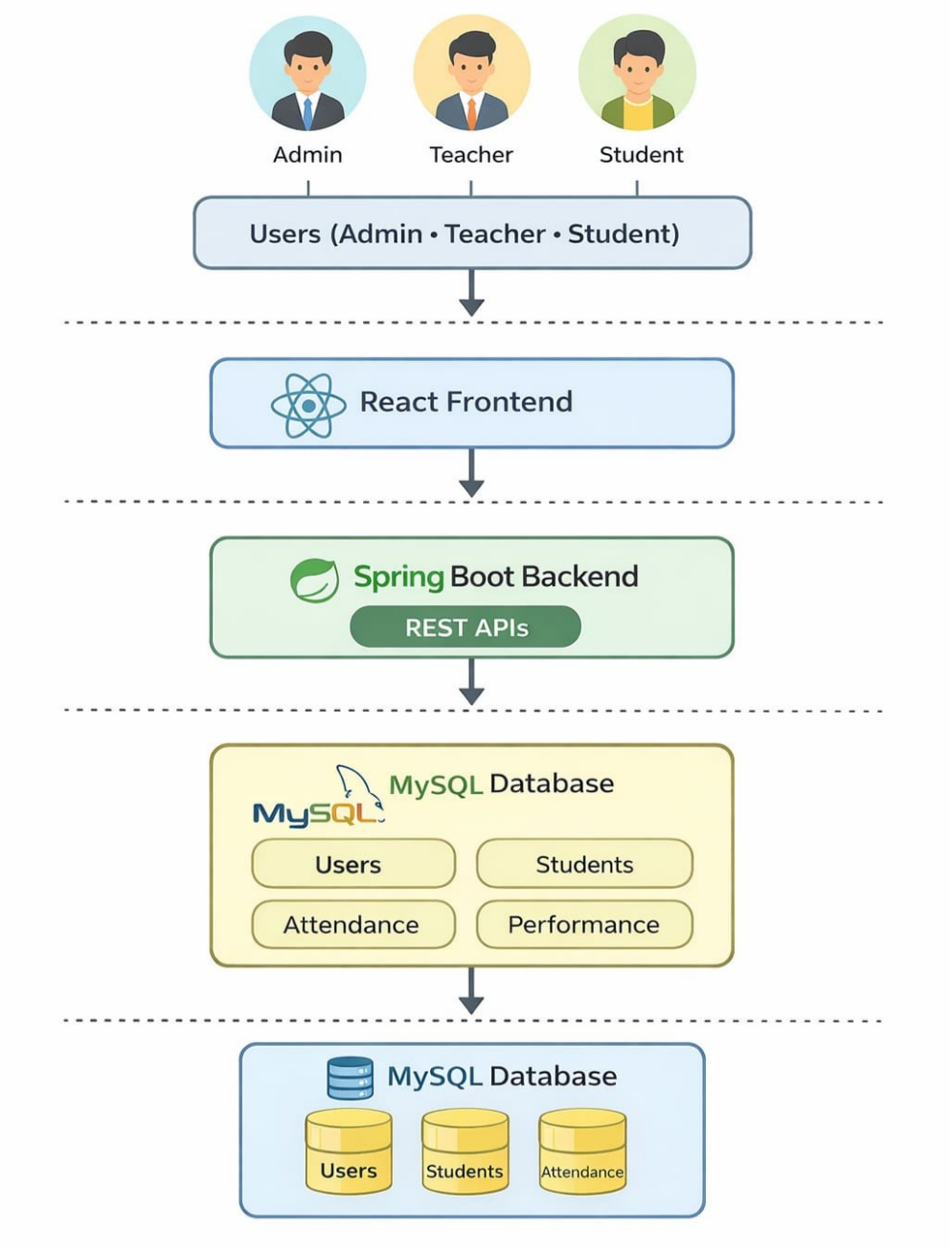
Backend Layer

• The backend is implemented using Spring Boot.
• It manages business logic, API endpoints, authentication, and data processing.
• The backend communicates with the database using REST APIs.

Database Layer

• MySQL is used as the database management system.
• It stores user information, student records, attendance data, and performance data.

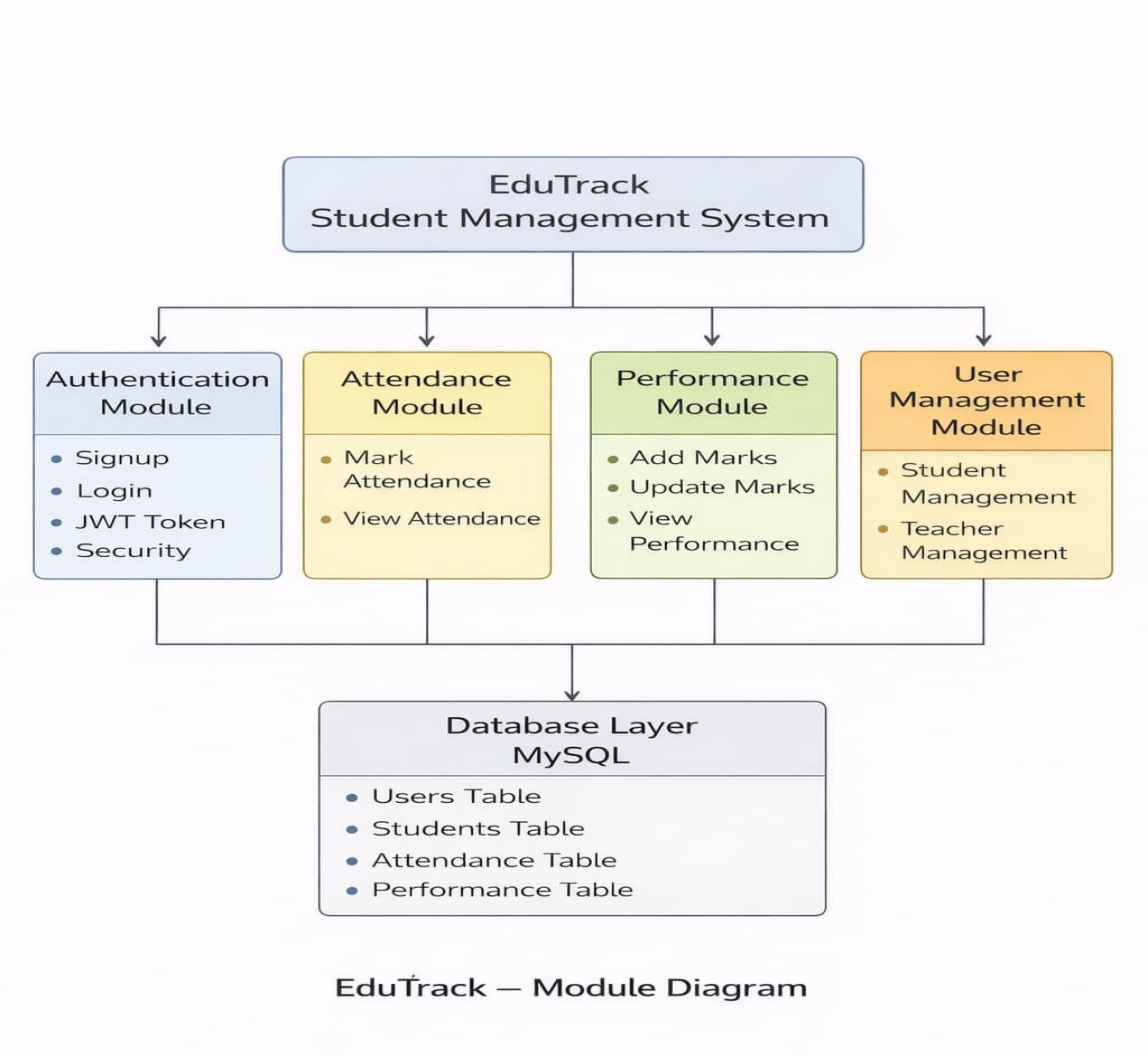
Figure 3.1: System Architecture Diagram of the EduTrack System



4. System Modules

The EduTrack platform is divided into several modules for better organization and maintainability.

Figure 4.1: EduTrack System Module Diagram



4.1 Authentication Module

This module manages user authentication and system access.

Functions include:

• User registration
• User login
• Authentication validation

4.2 User Management Module

This module manages user information.

Functions include:

• Managing administrator and teacher accounts
• Updating user profiles
• Managing user access permissions

4.3 Student Management Module

This module stores and manages student information.

Functions include:

• Adding new students
• Updating student details
• Deleting student records
• Viewing student information

4.4 Attendance Management Module

This module handles recording and tracking of student attendance.

Functions include:

• Marking daily attendance
• Viewing attendance history
• Calculating attendance percentage

4.5 Performance Management Module

This module manages student academic performance.

Functions include:

• Adding marks or performance scores
• Updating performance records
• Viewing performance reports

4.6 Analytics Module

This module provides insights into student performance and attendance trends.

Features include:

• Attendance percentage calculation
• Student performance statistics
• Summary reports and analytics

---

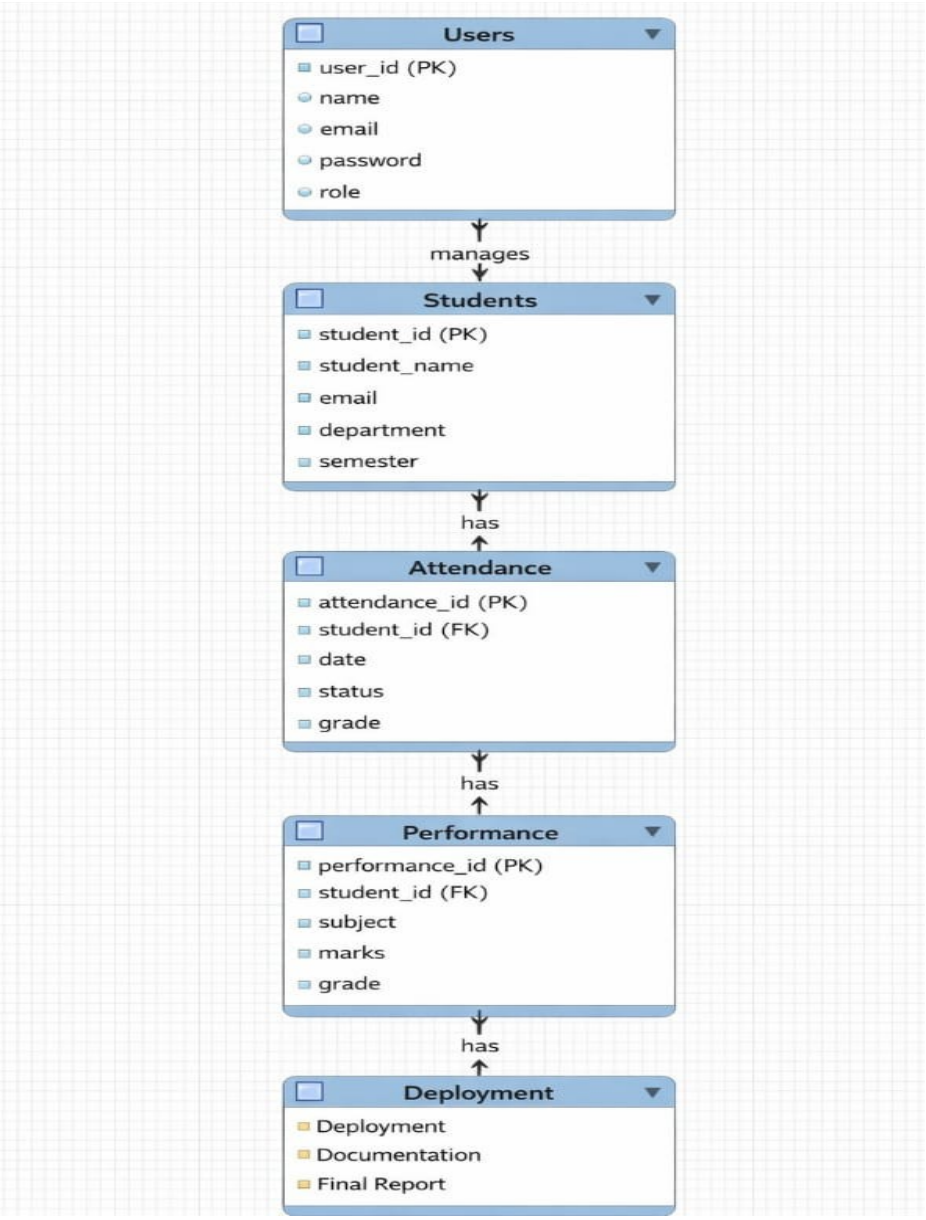
5. Database Design

The EduTrack system database consists of the following main tables.

Main Tables

- Users
- Students
- Attendance
- Performance

Figure 5.1: Entity Relationship Diagram of EduTrack Database

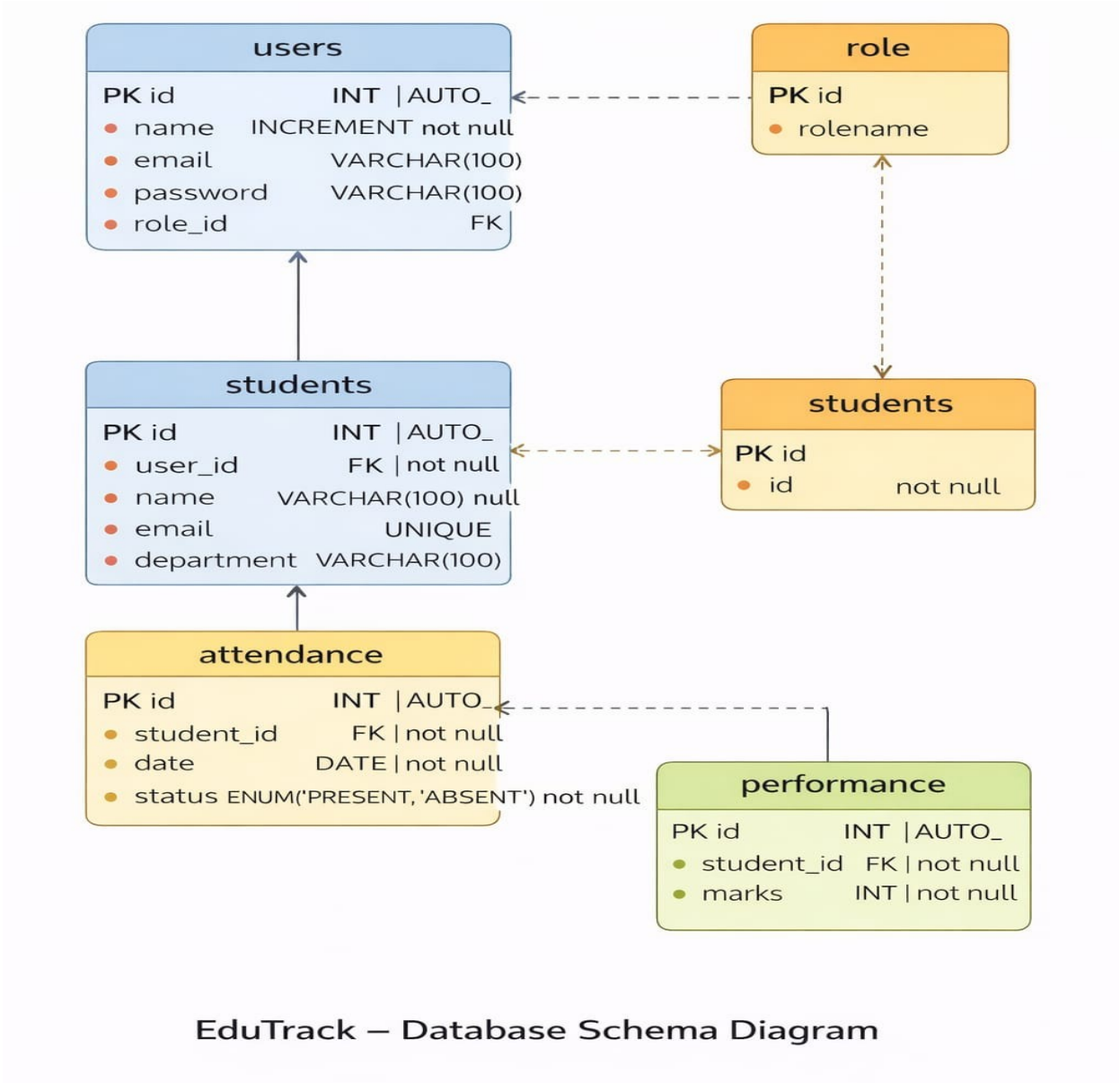


The Relationship Overview :

The EduTrack database maintains relationships between users, students, attendance records, and performance data. A user such as an administrator or teacher manages multiple student records in the system. Each student can have multiple attendance entries and performance records, which allows the system to track academic progress and attendance history effectively.

```
Users
│
└── Students
    │
    ├── Attendance
    │
    └── Performance
```

Figure 5.2: Database Schema Diagram for EduTrack System



Explanation

• A user (teacher or administrator) manages student records.
• Each student can have multiple attendance records.
• Each student can also have multiple performance records.

---

6. Technology Stack

The EduTrack system uses the following technologies:

Layer	Technology
Frontend	React.js
Backend	Spring Boot
Database	MySQL
Security	JWT Authentication
ORM	Hibernate / JPA
Build Tool	Maven

This technology stack enables scalable and secure system development.

---

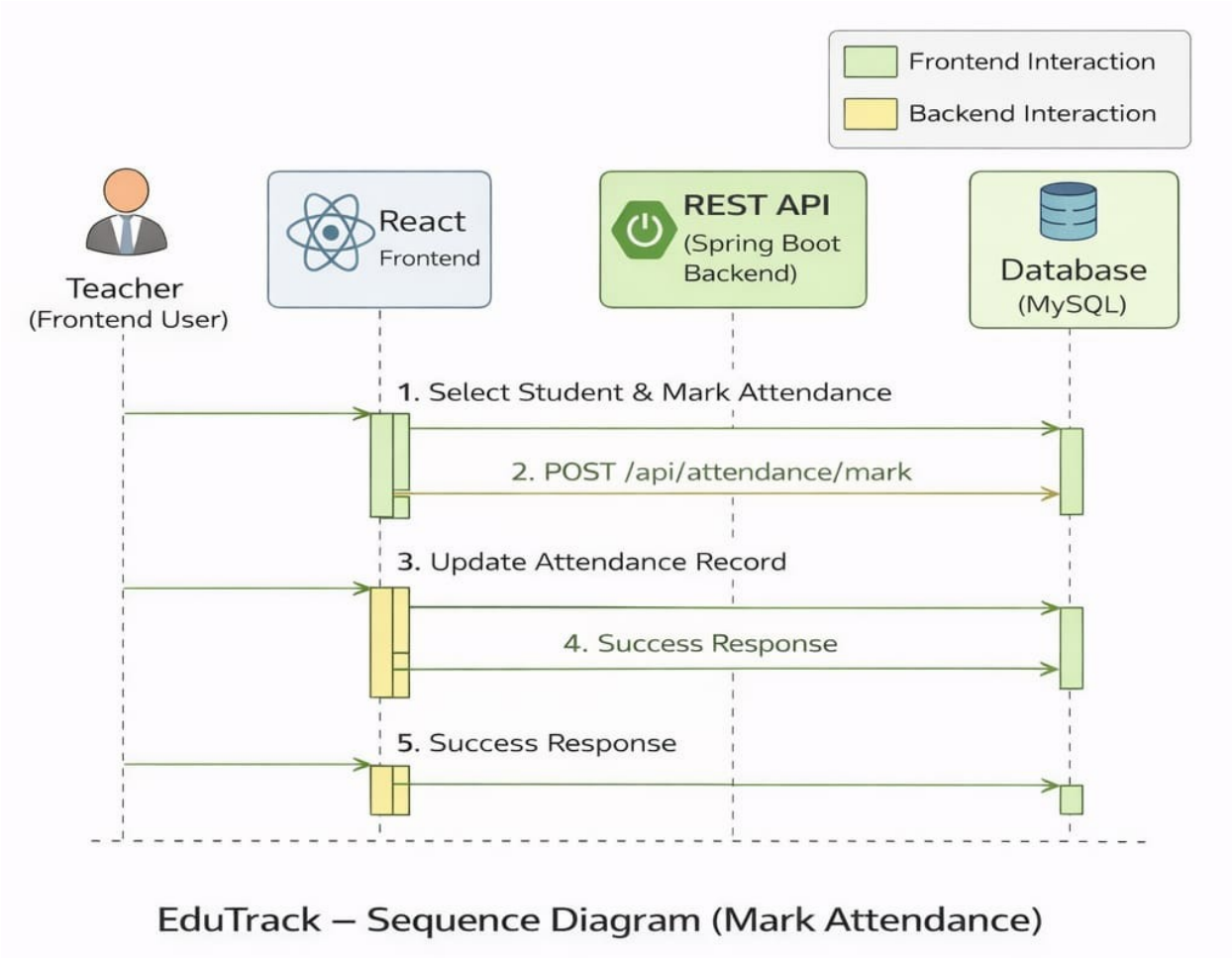
7. System Workflow

This section describes the sequence of steps and interactions between users and the EduTrack system for performing key operations such as recording attendance and managing student performance.

Example Workflow: Recording Attendance

Teacher Login
↓
View Student List
↓
Select Student
↓
Mark Attendance
↓
Backend Processes Request
↓
Attendance Record Stored in Database
↓
Attendance Updated Successfully

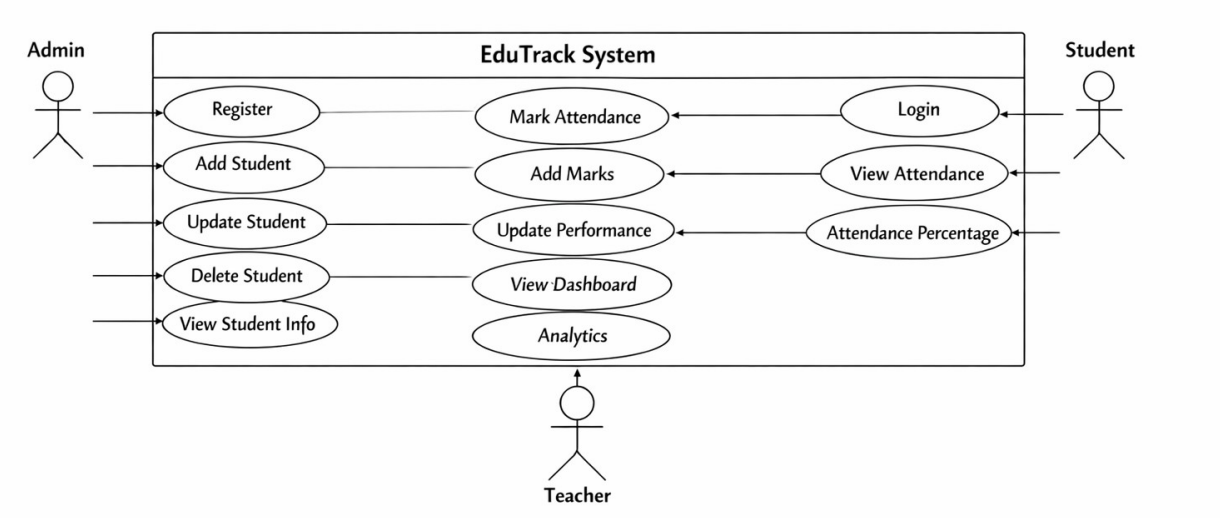
Figure 7.1: Sequence Diagram for Recording Attendance



7.2 Use Case Diagram

- *The Use Case Diagram illustrates the interactions between system users and the EduTrack system, showing how different actors such as administrators, teachers, and students access system functionalities like managing student records, recording attendance, and viewing performance reports.*
- This diagram below represents the system behavior from the user's perspective and highlights the main services offered by the EduTrack platform.
- *This diagram ensures that all important user interactions with the system are properly identified and documented during the design phase.*
- *It serves as a foundation for designing other system diagrams such as sequence diagrams and activity diagrams.*

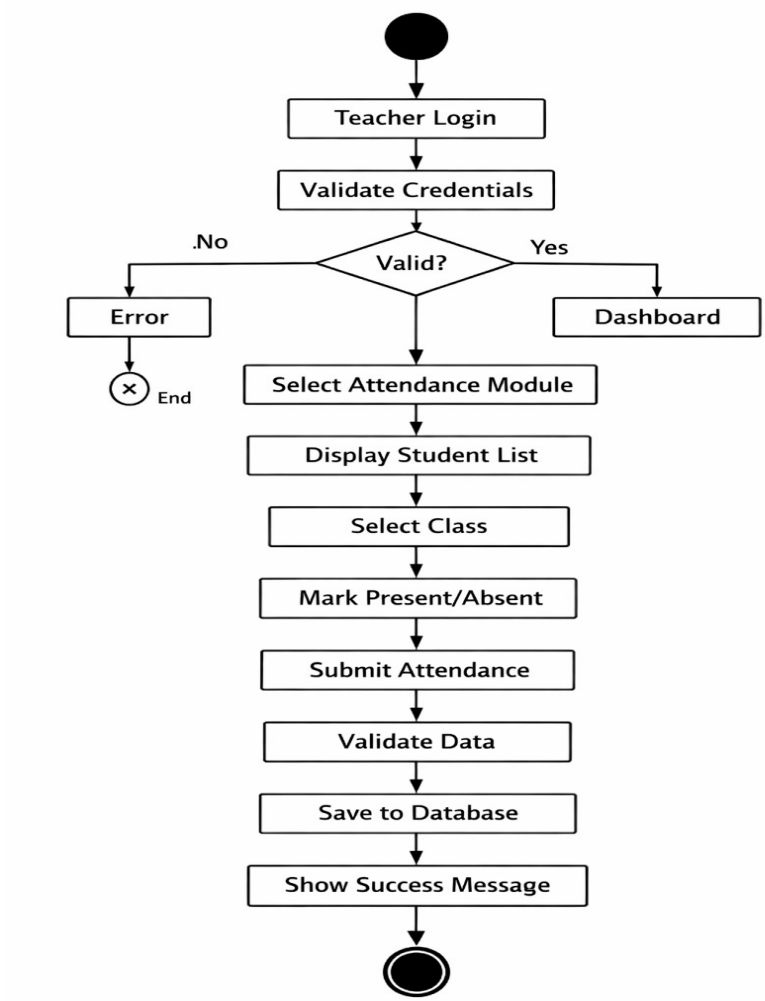
Figure 7.2 – Use Case Diagram



7.3 Activity Diagram

The Activity Diagram illustrates the workflow of the EduTrack system by showing the sequence of activities involved in performing different operations.

Figure 8.3 – Activity Diagram



8. Risk Analysis and Mitigation

Risk	Mitigation
Unauthorized system access	Implement authentication and role-based access control
Data inconsistency	Use database constraints and validation checks
System performance issues	Optimize queries and API performance
Data loss	Implement database backup and recovery

---

9. Testing Strategy

Testing ensures system reliability and functionality.

Unit Testing

• Testing individual system components such as services and controllers.

Integration Testing

• Testing interaction between modules and database.

System Testing

• Testing the entire application to verify system functionality.

User Acceptance Testing

• Testing the system with users to confirm requirements are met.

---

10. Success Criteria

The EduTrack system will be considered successful if:

• Student records can be managed successfully.
• Attendance records are stored and retrieved accurately.
• Student performance data is recorded and displayed correctly.
• The system operates securely and efficiently.
• Users can easily interact with the system through the interface.

---

11. Appendices

The appendices section may include additional technical documentation such as:

• API documentation
• Database schema diagrams
• System architecture diagrams
• Technical references